Worksheet 3

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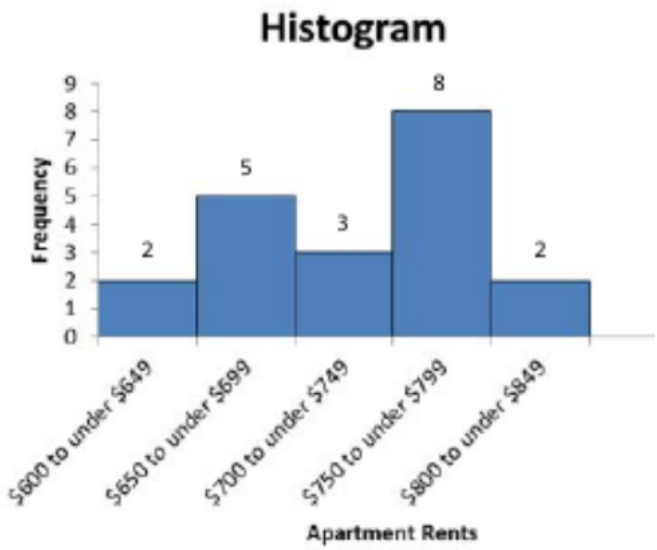
Remaining from Worksheet 2:

2. The following histogram shows the distribution of the monthly rental for a random sample of one-bedroom apartments in York, Pennsylvania.

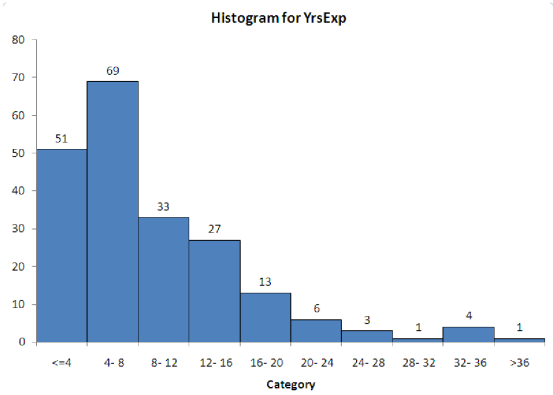


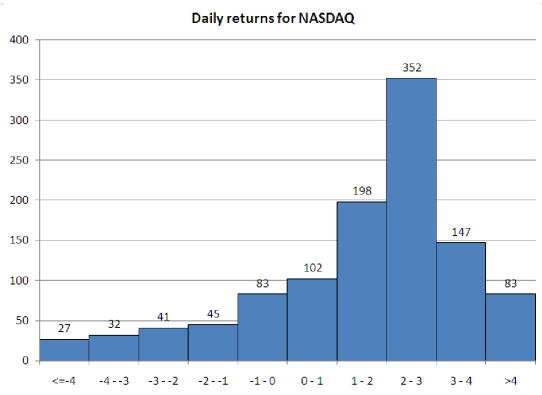
   a. What is the total number of apartments in this sample, and what is the percentage of monthly rents that are $750 and above? If we rank the observations from low to high, what can you say about the range of 8th ranked observation?

   b. Suppose the rent for one of the apartments in the sample is $680. How many apartments rent in the sample are going to be lower than or equal to this rent number?

   c. Identify the shape of below histograms [1]

[1] Graphs are from http://citadel.sjfc.edu/faculty/kgreen/MSTI130/MSTI130Text/Text_Fall_2014su28.html


Histogram for YrsExp
80
70
60
50
40
30
20
10
0
51
69
33
27
13
6
3
1
4
1
<=4
4- 8
8- 12
12- 16
16- 20
20- 24
24- 28
28- 32
32- 36
>36
Category


Daily returns for NASDAQ
400
350
300
250
200
150
100
50
0
27
32
41
45
83
102
198
352
147
83
<=-4
-4 - -3
-3 - -2
-2 - -1
-1 - 0
0 - 1
1 - 2
2 - 3
3 - 4
>4

3. The U.S. National Debt over the span of a decade from 1991 to 2001 is given in the following table:

Year	Debt (in T)
1991	7.3
1992	7.9
1993	8.3
1994	8.6
1995	8.9
1996	9.1
1997	9.2
1998	9.3
1999	9.2
2000	9.0
2001	8.9

a. Is this an example of time series data or cross sectional data?

b. Make an appropriate plot for this data.

c. What can you conclude from this data?

4. The following data has mean income and housing for 10 cities in Florida. Values are in dollars ($) and rounded to the nearest thousand.

City	Income ($\boldsymbol{x}$)	Housing (y)
A	26	109
B	29	97
C	25	115
D	28	99
E	38	122
F	32	145
G	25	100
H	22	76
I	29	113
J	42	144

a. Draw an appropriate diagram representing the relationship between Income (x) and Housing (y).

b. How would you describe the relationship between the income (x) and housing (y)?

5. A random sample of 10 students were asked how far they live from UMBC. These were their answers, rounded to the nearest half-mile:

$$1, 2.5, 1.5, 2, 0, 5, 3, 3.5, 1, 0.5$$

Compute the mean, median and mode for this.